Assignment - 1 (MAT 216)

Farhan Sadik

24101406

Sec - 1X

Ans. to the que. no. 1

(a)

$$x + 3z + 6t = 0$$

$$y + 2z + 5t = 0$$

But, the system has two ~~function~~ equations and four variables, so,

1st equ. $x = -3z - 6t$

2nd equ. $y = -2z - 5t$

Let, $z = s$ and $t = u$ (free parameters), so, the solution,

$$\begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = s \begin{bmatrix} -3 \\ -2 \\ 1 \\ 0 \end{bmatrix} + u \begin{bmatrix} -6 \\ -5 \\ 0 \\ 1 \end{bmatrix} \qquad [s, u \in \mathbb{R}]$$

(b) A particular solution is found by setting $z = 0$, $t = 0$.

$\therefore$ so, $x = 9$ $\qquad [x + 0 + 0 = 9]$

$y = 7$ $\qquad [y + 0 + 0 = 7]$

The homogenous solution is the same as in (a).

Thus the general solution is,

$$\begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 9 \\ 7 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -3 \\ -2 \\ 1 \\ 0 \end{bmatrix} + u \begin{bmatrix} -6 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \quad s, u \in \mathbb{R}$$

(c) For, $A\vec{x} = \vec{0}$ and $A\vec{x} = \vec{b}$ with $\vec{b} = \begin{pmatrix} 9 \\ 7 \end{pmatrix}$

The solution set of $A\vec{x} = \vec{b}$ is the set of all vectors of the form $\vec{x_p} + \vec{x_h}$, where $\vec{x_p}$ is a particular solution to $A\vec{x} = \vec{b}$ and $\vec{x_h}$ is any solution to $A\vec{x} = \vec{0}$. It is an affine space.

Ans to the que. no. 2

System,

$$3x+5y-2z+7t=4$$
$$-6x+y+3z-2t=-1$$
$$9x-4y-5z+8t=7$$

Augmented Matrix,

$$\left(\begin{array}{cccc|c} 3 & 5 & -2 & 7 & 4 \\ -6 & 1 & 3 & -2 & -1 \\ 9 & -4 & -5 & 8 & 7 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 5/3 & -2/3 & 7/3 & 4/3 \\ -6 & 1 & 3 & -2 & -1 \\ 9 & -4 & -5 & 8 & 7 \end{array}\right) \quad \left[R_1' = R_1/3\right]$$

$$\left(\begin{array}{cccc|c} 1 & 5/3 & -2/3 & 7/3 & 4/3 \\ 0 & 11 & -1 & 12 & 7 \\ 0 & -19 & 1 & -13 & -5 \end{array}\right) \quad \begin{array}{l} \left[R_2' = R_2 + 6R_1\right] \\ \left[R_3' = R_3 - 9R_1\right] \end{array}$$

$$\left(\begin{array}{cccc|c} 1 & 5/3 & -2/3 & 7/3 & 4/3 \\ 0 & 1 & -1/11 & 12/11 & 7/11 \\ 0 & 0 & -8/11 & 85/11 & 78/11 \end{array}\right) \quad \begin{array}{l} [R_2' = R_2/11] \\ [R_3' = R_3 + 19R_2] \end{array}$$

$$\left(\begin{array}{cccc|c} 1 & 5/3 & -2/3 & 7/3 & 4/3 \\ 0 & 1 & -1/11 & 12/11 & 7/11 \\ 0 & 0 & 1 & -85/8 & -39/4 \end{array}\right) \quad [R_3' = R_3 * (-11/8)]$$

$$\left(\begin{array}{cccc|c} 1 & 5/3 & 0 & -19/4 & -31/6 \\ 0 & 1 & 0 & 1/8 & -1/4 \\ 0 & 0 & 1 & -85/8 & -39/4 \end{array}\right) \quad \begin{array}{l} [R_2' = R_2 + R_3/4] \\ [R_1' = R_1 + \frac{2R_3}{3}] \end{array}$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & -119/24 & -68/12 \\ 0 & 1 & 0 & 1/8 & -1/4 \\ 0 & 0 & 1 & -85/8 & -39/4 \end{array}\right) \quad [R_1' = R_1 - 5/3 R_2]$$

Pivot columns: 1, 2, 3 ~~cols~~ (x, y, z)

Free Variable: 4 column

in parametic form, Let $t = s$ (free parameter)

$$x = -\frac{19}{4} + \frac{119}{24}s$$

$$y = -\frac{1}{4} - \frac{1}{8}s$$

$$z = -\frac{39}{4} + \frac{85}{8}s$$

$$t = s$$

Vector form,

$$\begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} -\frac{19}{4} \\ -\frac{1}{4} \\ -\frac{39}{4} \\ 0 \end{pmatrix} + s\begin{pmatrix} \frac{119}{4} \\ -\frac{1}{8} \\ \frac{85}{8} \\ 1 \end{pmatrix} \qquad s \in \mathbb{R}$$

Ans. to the que. no. 3

Augmented Matrix, $[A|I]$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 6 & 0 & 1 & 0 \\ 2 & 5 & 7 & 0 & 0 & 1 \end{array}\right)$$

Row operations,

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ \cancel{2} & \cancel{5} & \cancel{7} & & & \\ 0 & 1 & 1 & -2 & 0 & 1 \end{array}\right) \quad \begin{array}{l} [R_2' = R_2 - 2R_1] \\ [R_3' = R_3 - 2R_1] \end{array}$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array}\right) \quad [R_3' = R_3 - R_2]$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 5 & -2 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array}\right) \quad [R_1' = R_1 - 2R_2]$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 5 & 1 & -3 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array}\right) \quad [R_1' = R_1 - 3R_3]$$

Inverse is,

$$\begin{pmatrix} 5 & 1 & -3 \\ -2 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

Elementary Matrices from row operations applied to I_3

E_1 $(R_2' = R_2 - 2R_1)$

$$E_1 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

E_2 $(R_3' = R_3 - 2R_1)$

$$E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$$

E_3 $(R_3' = R_3 - R_2)$

$$E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

E_4 $(R_1' = R_1 - 2R_2)$

$$E_4 = \begin{pmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

E_5 $(R_1' = R_1 - 3R_3)$

$$E_5 = \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$\therefore$ $E_5 E_4 E_3 E_2 E_1 A = I_3$

The inverse is $A^{-1} = E_5 E_4 E_3 E_2 E_1$ (since $A^{-1} = E_5 E_4 E_3 E_2 E_1 I_3$)

Each elementary matrix corresponds to a row operation that reduces A to I_3. The product

$E_k \cdots E_1 A = I_3$ Implies $A^{-1} = E_k \cdots E_1$

Ans. to the que. no. 4

① Given system,

$$\left(\begin{array}{cc|c} 3 & -4 & k \\ -6 & 8 & 5 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 3 & -4 & k \\ 0 & 0 & 5+2k \end{array}\right) [R_2' = R_2 + 2R_1]$$

So, if $5+2k = 0$ then the system will be consistent

$\therefore k = -\frac{5}{2}$

$$\left(\begin{array}{cc|c} k & 1 & -2 \\ 4 & -1 & 2 \end{array}\right)$$

for all k the system will be consistent and when $k = -4 \Rightarrow$ infinitely many solution, else unique solution.

(b) Given reduced system,

$$x+y+z = 6$$
$$y+2z = 4$$
$$(\lambda-3)z = \mu-10$$

(i) No solution : $\lambda=3$ and $\mu\neq10$

(ii) Many solution: $\lambda=3$ and $\mu=10$

(iii) Unique Solution: $\lambda\neq3$

Ans. to the que. no. 5

Given System,

$$x+2y = 2$$
$$2x+y+z = 1$$
$$2x+z = 2$$

Augmented Matrix

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 2 \\ 2 & 1 & 1 & 1 \\ 2 & 0 & 1 & 2 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 2 \\ 0 & -3 & 1 & -3 \\ 0 & -4 & 1 & -2 \end{array}\right) \quad \begin{array}{l} R_2' = R_2 - 2R_1 \\ R_3' = R_3 - 2R_1 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 2 \\ 0 & -3 & 1 & -3 \\ 0 & 0 & -1 & 6 \end{array}\right) \quad R_3' = 3R_3 - 4R_2$$

Now, $\therefore z = -6$

$-3y = -3$

$\therefore y = -1$

$x = 4$

$\therefore \quad x = 4, \quad y = -1, \quad z = -6$

Ans.

Ans. to the que. no. 6

Given system,

$$3x + y + 3z = 2$$
$$x + 2z = -3$$
$$2x + y + z = 4$$

Augmented Matrix,

$$\left(\begin{array}{ccc|c} 3 & 1 & 3 & 2 \\ 1 & 0 & 2 & -3 \\ 2 & 1 & 1 & 4 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & -3 \\ 0 & 1 & -3 & 11 \\ 0 & 1 & -3 & 10 \end{array}\right)$$

$R_1 \leftrightarrow R_2$

$R_2' = R_2 - 3R_1$

$R_3' = R_3 - 2R_1$

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & -3 \\ 0 & 1 & -3 & 11 \\ 0 & 0 & 0 & -1 \end{array}\right)$$

$R_3' = R_3 - R_2$

The ~~equations~~ system of equations have no solution.

Ans. to the que. no. ~~6~~ 7

~~Given~~

Augmented Matrix from given system,

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 2 & 1 \\ 0 & 2 & 2 & 1 & 0 \\ 2 & 0 & 2 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 2 & 1 \\ 0 & 2 & 2 & 1 & 0 \\ 0 & -2 & 0 & -3 & -1 \end{array}\right) \quad R_3' = R_3 - 2R_1$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 2 & 1 \\ 0 & 2 & 2 & 1 & 0 \\ 0 & 0 & 2 & -2 & -1 \end{array}\right) \quad R_3' = R_3 + R_2$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 2 & 1 \\ 0 & 2 & 2 & 1 & 0 \\ 0 & 0 & 1 & -1 & -\frac{1}{2} \end{array}\right) \quad R_3' = \frac{R_3}{2}$$

Let, $z = t$ (free variable) $t \in \mathbb{R}$

$\therefore y = -\frac{1}{2} + t$

$2x = -2\left(-\frac{1}{2} + t\right) - t$

$\therefore x = \frac{1 - 3t}{2}$

$\cancel{x = \frac{1}{2} - \frac{3}{2}t}$

$\therefore w = 1 - \frac{3}{2}t$

$$\therefore \begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{1}{2} \\ -\frac{1}{2} \\ 0 \end{pmatrix} + \begin{pmatrix} -\frac{3}{2} \\ -\frac{3}{2} \\ 1 \\ 1 \end{pmatrix} t \quad , t \in \mathbb{R}$$

Ans.

Ans. to the que. no. 8

Given Matrix,

$$\begin{pmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{pmatrix}$$

$$\begin{pmatrix} 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & 3 & -6 & 6 & 4 & -5 \\ 0 & -1 & 2 & -2 & -1 & 3 \end{pmatrix}$$

$R_1 \leftrightarrow R_2$

$R_3' = R_3 - R_1$

$R_3' = R_3/2$

$$\begin{pmatrix} 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & -1 & 2 & -2 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{pmatrix}$$

$R_2 \leftrightarrow R_3$

$R_3' = R_3 + 3R_2$

$$\begin{pmatrix} 3 & 0 & -6 & 9 & 15 & -12 \\ 0 & 1 & -2 & 2 & 0 & -2 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{pmatrix}$$

$R_2' = -R_2$

$R_1' = R_1 + 7R_2$

$R_2' = R_2 - R_3$

$$\begin{pmatrix} 1 & 0 & -2 & 3 & 0 & -24 \\ 0 & 1 & -2 & 2 & 0 & -2 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{pmatrix}$$

$R_1' = R_1 - 15R_3$

$R_1' = R_1/3$

Pivot columns : 1, 2, 5

Free variables : 3, 4, 6 (Column)

Ans

Ans. to the que. no. 9

System,

$$3x_1 + 2x_2 + x_3 = 5$$

$$2x_1 - x_2 + 3x_3 = 2$$

$$x_1 + 3x_2 - x_3 = 3$$

Augmented Matrix,

$$\left(\begin{array}{ccc|c} 3 & 2 & 1 & 5 \\ 2 & -1 & 3 & 2 \\ 1 & 3 & -1 & 3 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 3 & -1 & 3 \\ 0 & -7 & 5 & -4 \\ 0 & -7 & 4 & -4 \end{array}\right)$$

$R_1 \leftrightarrow R_2$

$R_2' = R_2 - 2R_1$

$R_3' = R_3 - 3R_1$

$$\left(\begin{array}{ccc|c} 1 & 3 & -1 & 3 \\ 0 & -7 & 5 & -4 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$R_3' = R_3 - R_2$

$R_3' = -R_3$

$$\left(\begin{array}{ccc|c} 1 & 3 & 0 & 3 \\ 0 & -7 & 0 & -4 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$R_2' = R_2 - 5R_3$

$R_1' = R_1 + R_3$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 9/7 \\ 0 & 1 & 0 & 4/7 \\ 0 & 0 & 1 & 0 \end{array}\right) \qquad R_2' = -R_2/7$$

$$R_1' = R_1 - 3R_2$$

$\therefore\ x_3 = 0\ ,\ x_2 = 4/7,\ x_1 = 9/7$

$$\therefore \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 9/7 \\ 4/7 \\ 0 \end{pmatrix}$$

Ans.

Ans. to the que. no. 10

Inverse of A,

$[A \mid I]$

$$\left(\begin{array}{ccc|ccc} 2 & -1 & 4 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 3 & -2 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & -3 & 2 & 1 & -2 & 0 \\ 0 & -5 & -2 & 0 & -3 & 1 \end{array}\right)$$

$R_1 \leftrightarrow R_2$

$R_2' = R_2 - 2R_1$

$R_3' = R_3 - 3R_1$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & -3 & 2 & 1 & -2 & 0 \\ 0 & 0 & 1 & \frac{5}{16} & -\frac{1}{16} & -\frac{3}{16} \end{array}\right)$$

$R_3' = 3R_3 - 5R_2$

$R_3' = \frac{R_3}{(-16)}$

$$\left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{3}{16} & \frac{7}{16} & \frac{5}{16} \\ 0 & 1 & 0 & -\frac{2}{16} & \frac{10}{16} & -\frac{2}{16} \\ 0 & 0 & 1 & \frac{5}{16} & -\frac{1}{16} & -\frac{3}{16} \end{array}\right)$$

$R_2' = R_2 - 2R_3$

$R_1' = R_1 - R_3$

$\cancel{R_1' = 3R_1}$

$R_2' = -\frac{R_2}{3}$

$R_1' = R_1 - R_2$

$$\therefore \quad A^{-1} = \frac{1}{16}\begin{pmatrix} -3 & 7 & 5 \\ -2 & 10 & -2 \\ 5 & -1 & -3 \end{pmatrix}$$

Ans

Ans. to the que. no. 11

From given vectors,

$$x v_1 + y v_2 = b$$

System,

$$3x - 5y = 0$$
$$-2x + 6y = 4$$
$$x + y = 4$$

Augmented Matrix,

$$\left(\begin{array}{cc|c} 3 & -5 & 0 \\ -2 & 6 & 4 \\ 1 & 1 & 4 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 1 & 4 \\ 0 & 8 & 12 \\ 0 & -8 & -12 \end{array}\right) \quad \begin{array}{l} R_1 \leftrightarrow R_3 \\ R_2' = R_2' + 2R_1 \\ R_3' = R_3 - 3R_1 \end{array} \quad \left(\begin{array}{cc|c} 1 & 1 & 4 \\ 0 & 8 & 12 \\ 0 & 0 & 0 \end{array}\right) \quad R_3' = R_3 + R_2$$

So, the system ~~have~~ is consistent, ~~so that~~ because it has a unique solution.

So, vector b in the span of the vectors v_1, v_2

Ans.

Ans. to the que. no. 12

Augmented Matrix from given system,

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 3 & -1 & 2 & 3 \\ 4 & 1 & a^2-8 & a+3 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -6 \\ 0 & -7 & a^2-4 & a-9 \end{array}\right) \quad \begin{array}{l} R_2' = R_2 - 3R_1 \\ R_3' = R_3 - 4R_1 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -6 \\ 0 & 0 & (a^2-9) & (a-3) \end{array}\right) \quad R_3' = R_3 - R_2$$

No solution: $a = -3$

~~Many~~ Infinitely Many Solution: $a = 3$

Unique Solution: $a \neq 3$ and $a \neq -3$

Ans.

Ans. to the que. no. 13

(i) False : all given vectors have z-component 0, but b has z have 3

(ii) True : The solution set is the same by definition

(iii) False : If inconstinent, b is not in the span

(iv) False : The span contains all linear combinations, not just the vectors and zero.

Ans. to the que. no. 14

(i) 4X6 matrix : Largest number of pivots 4 (min no. of column)

(ii) 6X4 matrix : Largest number of pivots 4

(iii) 3 eq, 4unknowns : Infinitely many solution (at least one free variable)

(iv) Augmented matrix has 4 pivot columns (pivot in the augmented column)

Bonus Question

Ans. to the que. no. 1

(a)

Given,

$$s_1 = -\frac{3}{2} - 2t$$

$$s_2 = \frac{3}{2} + t$$

$$s_3 = t$$

Checking system,

(i) $s_1 + s_2 + s_3$

$$\Rightarrow \left(-\frac{3}{2} - 2t\right) + \left(\frac{3}{2} + t\right) + t$$

$$\Rightarrow 0$$

(ii) $s_1 + 3s_2 - s_3$

$$\Rightarrow \left(-\frac{3}{2} - 2t\right) + 3\left(\frac{3}{2} + t\right) - t$$

$$\Rightarrow 3$$

(i) $\neq$ (ii)

Not a solution for any t.

(b) General solution,

$$x = \alpha v_1 + \beta v_2 + \gamma v_3 + \delta v_4 + \lambda v_5$$

variables,

$$x_1 = 2\alpha + 3\beta + 2\gamma + 2\lambda$$

$$x_2 = -\lambda$$

$$x_3 = \delta$$

$$x_4 = 8\alpha + 4\beta + 3\gamma$$

$$x_5 = -\gamma$$

$$x_6 = -\beta$$

$$x_7 = 3\alpha$$

$$x_8 = -\alpha$$

Substitute into x_1 and x_4

$$x_1 = -2x_2 - 2x_5 - 3x_6 + \tfrac{7}{3}x_7$$

$$x_2 = -3x_5 - 4x_6 + \tfrac{8}{3}x_7$$

$$x_8 = -x_7$$

Now,

$$\begin{pmatrix} 1 & 2 & 0 & 0 & 2 & 3 & -7/3 & 0 \\ 0 & 0 & 0 & 1 & 3 & 4 & -8/3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

Pivot columns: 1, 4, 7